Jimmy Chen

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Education

Cornell University, College of Engineering — Ithaca, NY
B.S. in Computer Science, Minor in Artificial Intelligence | GPA: 3.43/4.0 — *Expected May 2028*

- **Relevant Coursework:** Analysis of Algorithms, Machine Learning, Natural Language Processing, Computer Architecture, Large-Scale ML Systems, Computer System Organization, OOP & Data Structures

Technical Skills

Languages: Python, C, Java, OCaml, Swift, JavaScript, SQL, RISC-V Assembly
AI/ML: PyTorch, TensorFlow, Core ML, scikit-learn, NumPy, CUDA, Transformers, NLP
Backend & Cloud: FastAPI, PostgreSQL, Supabase, REST APIs, AWS (EC2, S3, RDS, IAM)
Tools: Docker, Git, GitHub Actions, pytest, JUnit, SwiftUI, MapKit, HealthKit

Experience

Research Intern — May 2026 – Aug 2026
SEAL (Software Engineering and Analysis Lab), Cornell University — *Ithaca, NY*

- Reproduced TrainCheck's published evaluation on real-world silent-bug traces for the lab's PyMOP-LOLA extension, a runtime verification tool using LOLA's stream-based logic to catch silent bugs in ML training loops
- Detected violations in **7 of 10** supplied bug traces (9 of 13 curated invariants), including a genuine DeepSpeed bug later confirmed and fixed by maintainers
- Analyzed how to convert TrainCheck-mined training-trace invariants into LOLA monitor specifications, informing the extension's approach to spec generation

Research Assistant — Feb 2026 – Present
Smart Computer Interfaces for Future Interactions (SciFi) Lab, Cornell University — *Ithaca, NY*

- Built two SwiftUI iOS research apps for a mobile keyboard study, capturing 25+ per-keystroke features across 27+ keys plus IMU sensor data, and designed a per-key 2D Gaussian touch model to replace static key boundaries
- Engineered a second app using ReplayKit and a Broadcast Upload Extension (App Groups) for live screen recording, synchronizing camera, IMU, and keystroke data with automated cloud upload across 10 sessions
- Developed an automatic in-app analytics pipeline generating session summaries, tap-distribution visuals, and per-session Gaussian boundary reviews, reducing input latency by **73%**
- Built and shipped the on-device hand-grip classifier powering the app, a 1D-CNN over 50Hz IMU data (TensorFlow, Core ML) running live at 30Hz, reaching **∼95% accuracy** on unseen users (vs. 33% chance)

Web Developer Intern — Jun 2023 – Nov 2023
New York Tutoring Center — *New York City, NY*

- Launched production site for 100+ students within 6 months, increasing site traffic by **40%** and engagement by **60%** via SEO metadata, and managed user data with MySQL and Git-based version control

Projects

Seagull – 110M Parameter Transformer LLM | *Python, PyTorch* — Mar 2026 – Apr 2026

- Built and fine-tuned a 110M-parameter decoder-only transformer for caption generation, reducing perplexity from **121.9 to 22.6** on 79K New Yorker caption pairs after one epoch
- Implemented the full model's transformer architecture in PyTorch, including RoPE embeddings, RMSNorm, masked multi-head self-attention, SwiGLU feed-forward layers, and pre-norm residual transformer blocks

Cameldew Valley – 2D Farming Simulator | *OCaml, Raylib, SQLite3, MVC* — Nov 2025 – Dec 2025

- Collaborated on a 4-person Agile team to build an OCaml farming simulator using an MVC design, implementing the core game-state and pause systems across a **5000**+ line, **91**-file codebase
- Implemented the core Game State system and global pause system, and led development of an automated test suite for models, controllers, and game state, reaching **80%+ code coverage**

Stridr – iOS Running Tracker | *Swift, SwiftUI, Supabase, MapKit, HealthKit* — May 2025 – Aug 2025

- Built a full-stack iOS running app with real-time GPS route tracking, pace/distance calculation, and magic-link authentication via Supabase
- Designed a Supabase-backed PostgreSQL schema to persist workout history and GeoJSON routes, and integrated HealthKit to sync workouts to Apple Health